

WEATHER: sMAPE at each forecast step				
Lookback = 24 steps, Forecast horizon = 72 steps				
Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	61.843	73.819	62.087	32.603
2.000	63.561	67.858	63.966	36.087
3.000	64.460	66.507	65.105	38.192
4.000	65.155	66.488	65.961	39.484
5.000	65.757	66.644	66.548	40.672
6.000	66.330	66.769	67.179	41.564
7.000	66.787	66.881	67.664	42.183
8.000	67.247	67.082	68.044	43.229
9.000	67.505	67.299	68.367	43.693
10.000	67.909	67.481	68.593	44.137
11.000	67.928	67.570	68.884	44.362
12.000	68.020	67.657	69.166	44.689
13.000	67.975	67.735	69.401	45.028
14.000	68.161	67.805	69.586	45.289
15.000	68.434	67.874	69.706	45.735
16.000	68.481	67.970	69.934	46.037
17.000	68.354	68.060	70.146	46.590
18.000	68.261	68.152	70.356	46.744
19.000	68.347	68.229	70.485	46.706
20.000	68.482	68.298	70.710	47.337
21.000	68.644	68.328	70.890	47.416
22.000	68.800	68.328	71.000	47.639
23.000	68.795	68.328	71.100	47.736
24.000	68.772	68.385	71.118	48.099
25.000	68.889	68.480	71.094	48.075
26.000	69.026	68.591	71.129	47.984
27.000	68.965	68.651	71.113	48.254
28.000	69.021	68.717	71.147	48.363
29.000	69.252	68.777	71.240	48.495
30.000	69.524	68.864	71.290	48.757
31.000	69.534	68.934	71.364	49.022
32.000	69.487	68.934	71.334	49.080
33.000	69.881	68.978	71.428	49.454
34.000	70.054	69.004	71.571	49.711
35.000	69.917	69.040	71.627	49.842
36.000	69.730	69.066	71.741	49.814
37.000	69.940	69.088	71.875	49.911
38.000	70.002	69.103	72.030	50.068
39.000	69.888	69.106	72.194	50.350
40.000	70.058	69.174	72.328	50.629
41.000	69.964	69.263	72.367	50.583
42.000	69.926	69.366	72.437	50.074
43.000	70.117	69.446	72.456	50.159
44.000	70.254	69.515	72.490	50.305
45.000	70.307	69.566	72.539	50.795
46.000	70.196	69.607	72.567	50.811
47.000	70.262	69.664	72.546	51.022
48.000	70.118	69.731	72.512	50.891
49.000	70.237	69.838	72.463	51.107
50.000	70.320	69.944	72.503	51.044
51.000	70.082	70.009	72.537	51.314
52.000	70.297	70.078	72.559	51.306
53.000	70.384	70.141	72.439	51.131
54.000	70.507	70.212	72.393	51.276
55.000	70.501	70.248	72.459	51.496
56.000	70.358	70.272	72.512	51.964
57.000	70.524	70.282	72.628	52.210
58.000	70.729	70.265	72.663	52.264
59.000	70.826	70.261	72.633	52.478
60.000	70.899	70.247	72.611	52.230
61.000	70.663	70.254	72.642	51.885
62.000	70.620	70.262	72.779	52.010
63.000	70.946	70.264	72.788	52.240
64.000	70.841	70.289	72.781	52.411
65.000	71.067	70.335	72.675	52.493
66.000	70.946	70.382	72.755	52.521
67.000	71.223	70.415	72.776	52.477
68.000	71.518	70.460	72.815	52.431
69.000	71.374	70.517	72.971	52.788
70.000	71.114	70.590	72.848	52.622
71.000	71.209	70.676	72.828	52.631
72.000	71.294	70.784	72.743	52.655

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. This is a percentage-style error that is easier to compare across scales.
• Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.